

flashjet

GPU-native jet clustering with the full merge history

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* speaker

- 1 Jet clustering, and flashjet
- 2 Correctness
- 3 Speed
- 4 Downstream: the clustering tree as tagger input

Jet clustering, and flashjet

Reconstruction is moving to accelerators — clustering has not

Reconstruction is going heterogeneous

- CMS runs GPUs in the HLT since Run 3
- **~30 %** of HLT reconstruction offloaded
- $\rightarrow$ **~25 %** less total HLT time

pixel local reco · pixel tracks + vertices · calorimeter local reco

Ported so far

- tracking · vertexing · calorimetry
- **not jet clustering**

Why it is still on the CPU

- sequential recombination is **IRC safe** — why it survived
- but each merge **depends on the previous one**
- that data dependence does not map onto a GPU in the obvious way

Where it hurts

- reclustering **inside a training loop**
- scanning the **jet radius**
- **substructure** on demand

each one pays a host $\leftrightarrow$ device transfer, over and over

An open-source GPU jet clusterer

- k_t , Cambridge–Aachen, anti- k_t
- custom **Triton** kernels

Why Triton

- the GPU language of the PyTorch ecosystem
- **compiles and autotunes** for the hardware at hand
- **no separate CUDA build**, no device code to maintain

Execution

- one program clusters **one event on chip**
- the batch goes out in **one fused launch**
- jets, histories, substructure → **tensors**

What makes it tractable

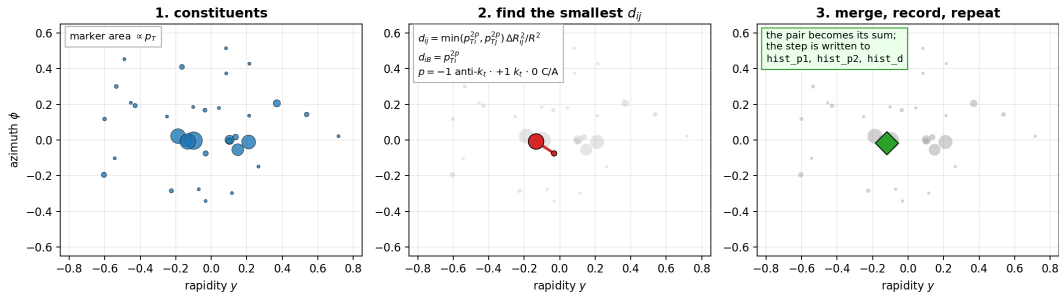
Cacciari–Salam NN lemma: the global d_{ij} minimum sits at some particle's **geometric** nearest neighbour.

- each slot carries its own NN
- a merge invalidates only $O(1)$ rows
- $\Rightarrow O(N^2)$, **not** $O(N^3)$

Engineering

- bandwidth-bound → NN update and **next argmin** fused into one sweep
- **branchless** pair-vs-beam (masked stores)
- autotuned **once per GPU model**, cached → **reproducible**

Sequential recombination — and what flashjet keeps from it



Repeat until nothing is left: if the smallest distance is a d_{ij} , merge that pair; if it is a d_{iB} , call that a jet and remove it. One exponent p selects the algorithm — the *same* kernel serves all three.

One exponent, three algorithms

The generalised- k_t distance measure:

$$d_{ij} = \min\left(p_{Ti}^{2p}, p_{Tj}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \quad \Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$$

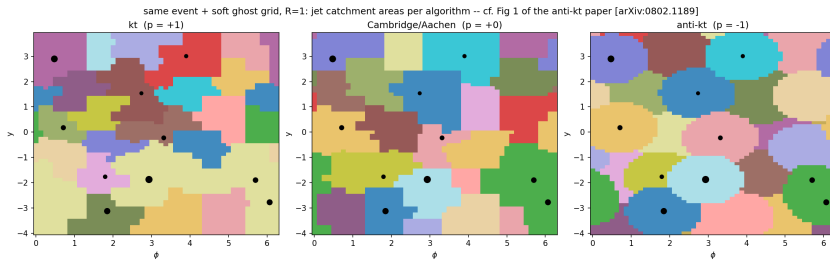
$$d_{iB} = p_{Ti}^{2p}$$

Merge the pair with the smallest d_{ij} ; if the smallest distance is a d_{iB} , call i a jet and remove it.

Only the exponent p changes:

p	algorithm	merges first	the tree is ordered by
-1	anti- k_t	around the hardest particles	— (gives the jets)
0	Cambridge–Aachen	the closest pair	angle
+1	k_t	the softest pair	relative p_T

The ordering is **what the substructure reads**: C/A is the tree for **grooming** and the **Lund plane**, k_t for **exclusive subjects**.



Jet catchment areas for the three algorithms, from flashjet's own clustering. k_t and C/A give ragged, fluctuating boundaries; **anti- k_t gives circles of area πR^2** around each hard particle — the defining result of the algorithm, recovered here without reference to FastJet.

It returns the tree, not just the jets

Clustering a batch returns **two** things — the jets, and the record of how they were built.

field	shape	what it holds
jet_idx	(B, N)	which jet each particle ended up in
n_jets	$(B,)$	how many jets were found
hist_p1, hist_p2	(B, N)	which pair merged , at each step
hist_child	(B, N)	what they merged <i>into</i>
hist_d	(B, N)	the distance d_{ij} at that step

Why the second half matters

The bottom three rows are the **complete binary merge tree**. Groomed mass, z_g , R_g , Lund coordinates and exclusive subjets are all **coordinates on that tree** — so they become array reads, with **no second clustering pass**.

Using it

```
import flashjet

out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
```

argument	type	
p4	$(B, N, 4)$ tensor	(p_x, p_y, p_z, E) ; CPU or CUDA
mask	(B, N) bool	marks real constituents in a padded batch
R	float	jet radius
algorithm	str	"antikt" "kt" "cambridge"
p	float	generalised- k_t exponent (overrides algorithm)

out.n_jets	# (B,)	jets found
out.jets_p4(p4)	# (B,J,4)	jet four-momenta
out.splitting_scales()	#	$\sqrt{d_{12}}$, $\sqrt{d_{23}}$, ...
out.exclusive_jets(n_jets=3)	# F1	exclusive k_t subjects
out.groomed_jets(p4, R, z_cut=0.1, beta=0.0)	# F2	soft drop / mMDT
out.lund_coordinates(p4, R)	# F3	primary Lund plane

The same call runs unchanged on CPU and GPU — the backend is chosen from the problem size, not by the caller.

What is implemented

Clustering

- generalised- k_t : anti- k_t , k_t , C/A
- batched, ragged input via a mask
- three backends, auto-dispatched by size
- merge history recorded in-kernel
- jet regime *and* event regime

Substructure — reads of the history

- **F1** exclusive k_t subjets
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

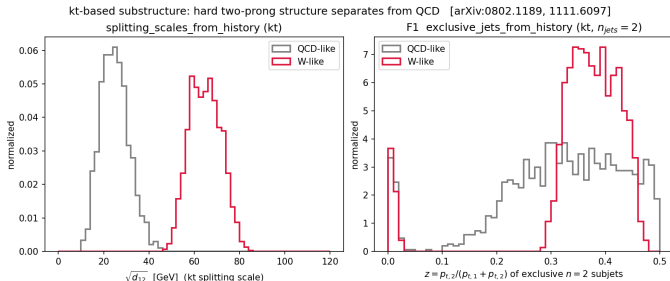
What it does. Undo the last merges of the sequence to expose exactly n subjects — or stop at a d_{cut} .

How. The history is already a binary tree. Walk back up it $n - 1$ steps; no reclustering.

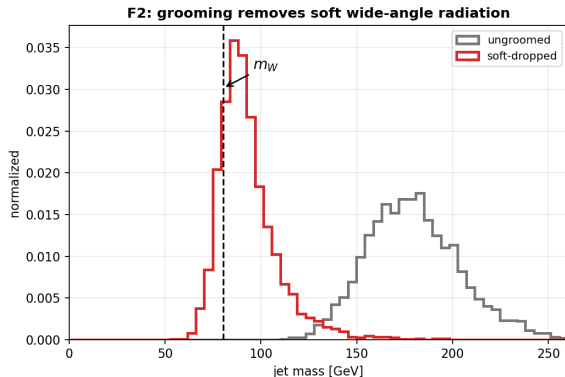
Why k_t . k_t merges the softest pair first, so the *last* merges are the hard prongs — exactly what a 2-prong decay leaves behind.

The splitting scales come straight from `hist_d`:

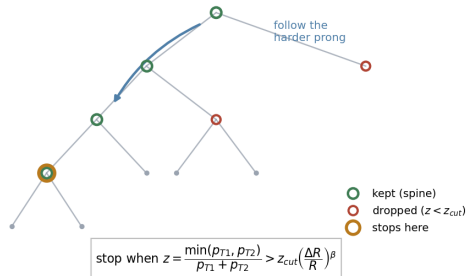
$$\sqrt{d_{12}}, \sqrt{d_{23}}, \dots \text{ with } d_{ij} = \min(p_{Ti}^2, p_{Tj}^2) \Delta R_{ij}^2 / R^2$$



Toy W-like (2-prong) vs QCD-like fat jets. **Left:** the k_t splitting scale $\sqrt{d_{12}}$ separates cleanly. **Right:** the momentum balance z of the two exclusive subjects — the W sits near $z \sim 0.4$, QCD spreads to small z .



the soft-drop walk on the C/A tree

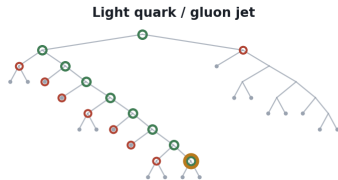


Walk down the C/A tree along the harder branch, discarding the softer prong until the momentum-sharing condition is met. $\beta = 0$ recovers mMDT.

The walk is $O(\log_2 n)$ on the stored tree — it never touches the constituents again, so grooming is essentially free once clustered.

C/A merge trees of real jets — root at top, constituents at the bottom

ATLAS Top Tagging Open Data · vertical = declustering depth · horizontal = angular ordering



a long spine: soft prong after soft prong
is stripped, and the mass collapses
 $n_{\text{drop}} = 8$ $m: 80 \rightarrow 1 \text{ GeV}$



the very first split is already balanced,
so nothing is groomed away
 $n_{\text{drop}} = 0$ $m: 80 \rightarrow 80 \text{ GeV}$



also stops at once, but on a wider split,
so the whole decay is kept
 $n_{\text{drop}} = 0$ $m: 170 \rightarrow 170 \text{ GeV}$

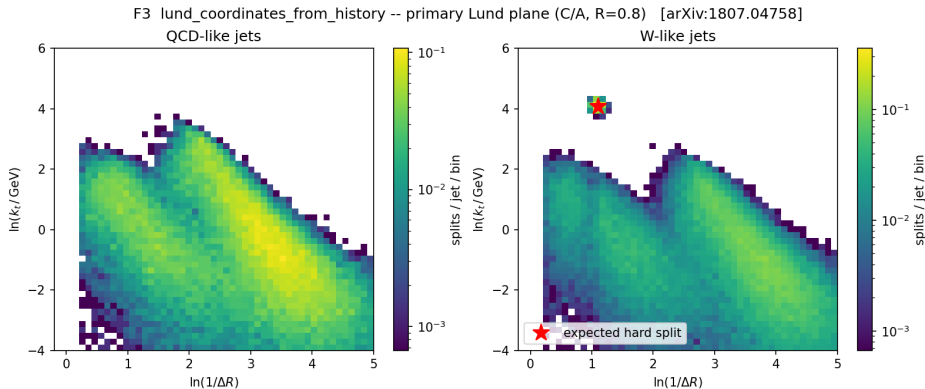
● soft-drop spine ● dropped prong ● grooming stops here ● constituent

A jet without a decay has nothing to stop on. The light-quark/gluon jet is a long *staircase*: soft prong after soft prong is stripped, and its mass collapses to nothing.

ATLAS Open Data, C/A with soft drop ($z_{\text{cut}}=0.1$, $\beta=0$). The number of drops is itself a discriminant — and it is read straight off the stored tree, at no extra cost.

A real decay stops at once. Both top jets halt on their very first split, because it is already hard and balanced — so the mass survives.

The number of drops is itself a discriminant — and it is read



What. Every primary splitting emits $z = \frac{p_{T2}}{p_{T1} + p_{T2}}$, ΔR , $k_t = p_{T2} \Delta R$, and the plane coordinates $(\ln \frac{1}{\Delta R}, \ln k_t)$.

QCD fills the soft-collinear region smoothly; the W-like sample adds a **localised hard splitting** exactly where the decay is predicted to sit (star).

How. One pass down the primary branch — each node is one point in the plane.

Check. The d -channel ties *exactly* to the splitting scales computed independently.

Correctness

Two kinds of validation

First: toys, where truth is known

On generated inputs the *answer* is known analytically — so any observable landing off the predicted curve is a real bug, with nothing to hide behind.

Input A — toy fat jets — QCD-like and W-like fat jets

Input B — toy parton shower — leading-log parton shower

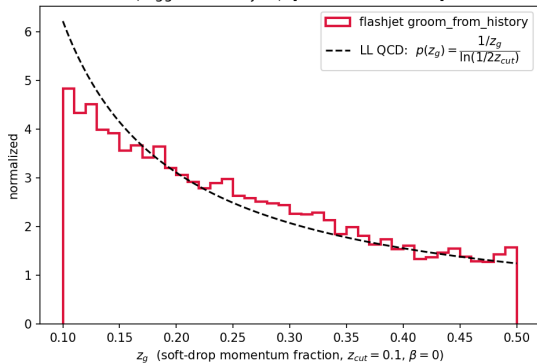
Then: real data, against FastJet

On detector-level jets there is no analytic answer — so the reference is **FastJet on the same events**, and the experiment's own reconstructed jets.

Input C — ATLAS Open Data — large- R jets and full events

The order matters: **toys first** pins the algorithm against a known answer; **real data second** shows it survives detector-level input. A failure at either stage means something different.

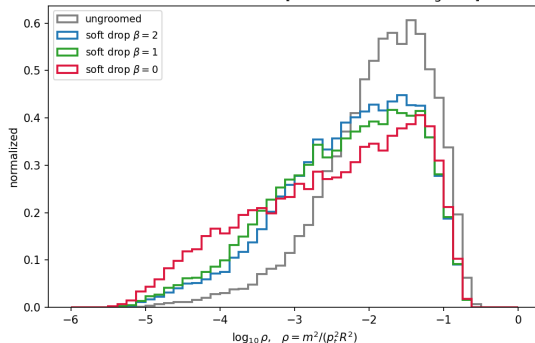
groomed splitting fraction vs the analytic prediction
(tagged 72% of jets) [cf. arXiv:1402.2657]



Soft-drop z_g follows the $1/z$ form predicted at leading-log, recovered from the C/A history.

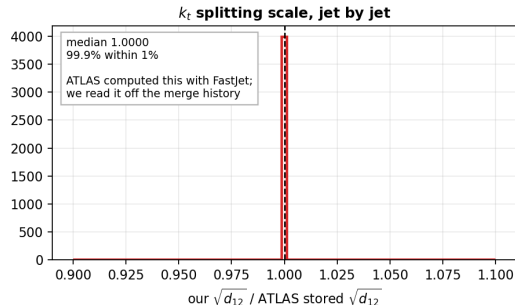
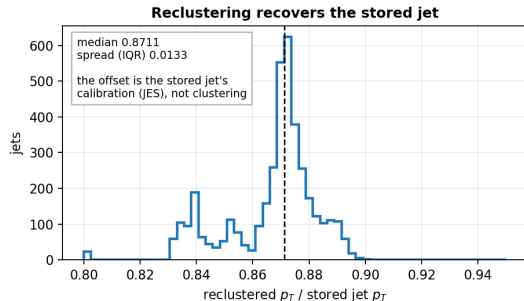
Toy-shower input, so the *prediction* is unambiguous — these test the decoders, not an event generator.

groomed jet mass: stronger grooming (smaller β) pushes the soft-radiation mass down [cf. arXiv:1402.2657 Figs 3-4]



Groomed-mass ordering in β : larger β grooms less, so the mass peak moves up.

Jet-by-jet closure against ATLAS-stored quantities



Left. Reclustering the stored constituents recovers the stored jet with a **0.013 spread**. The 0.87 offset is the stored jet's **calibration**, applied after clustering — it is present before we touch anything.

Right. The k_t splitting scale $\sqrt{d_{12}}$ against the value **ATLAS computed with FastJet** and stored: median **1.0000**, **99.9 %** within 1%. We read ours straight off the merge history.

3 algorithms $\times$ 5 radii $\times$ 5 multiplicity bins $\times$ 2 samples, 20 000 jets per bin, on ATLAS Open Data.

check	result
number of jets found	100.000 % at every one of 150 points
leading-jet p_T within 10^{-4}	1.0000 at 148/150 points
median relative difference	$\sim 7 \times 10^{-8}$ (float32 precision)

The grid is the point

A single working point proves little. Agreement holds across **all three algorithms**, **every radius** from 0.4 to 1.2, and **24 to 123** constituents per jet.

Two independent references

The same events are clustered by **FastJet** for the jets, and every substructure decoder is separately pinned against an **independent NumPy tree-walk**.

A stronger test than agreeing with FastJet.

ATLAS publishes an open dataset carrying both the **calorimeter clusters** and **its own reconstructed jets**. Cluster the ~ 600 clusters per event ourselves, and compare:

n_{jets} vs ATLAS	100.0 % match
mean jets/event	10.96 vs 10.96

This closes the algorithm against **the experiment's own published output** — not against another implementation of the same algorithm.

599.3 clusters/event: the event regime, where the problem is $O(N^2)$.

Speed

Data — ATLAS Open Data, freely presentable

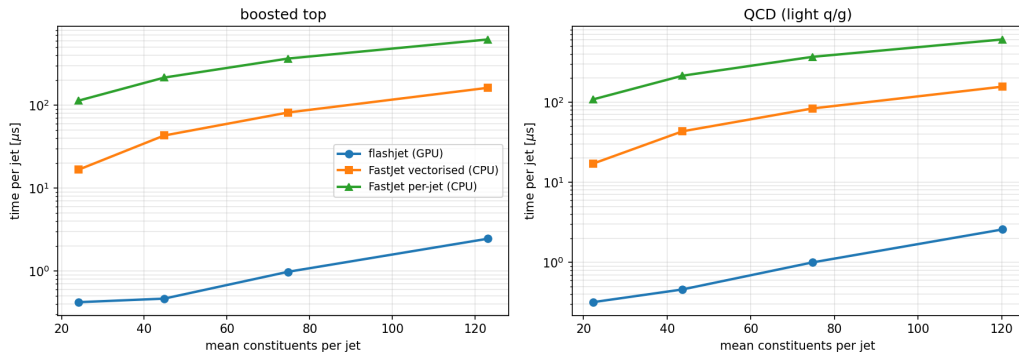
sample	jets	$\langle n_{\text{const}} \rangle$
boosted top	30 000	59.2
QCD (light q/g)	30 000	52.4

Large- R jets with calorimeter-cluster constituents. The grid is **3 algorithms** $\times$ **5 radii** $\times$ **5 multiplicity bins**, 20 000 jets per bin.

Method

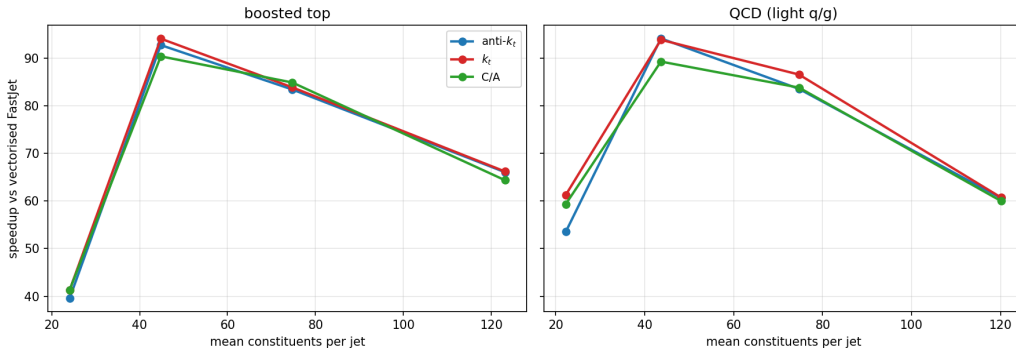
- 1 extract constituents once, save;
- 2 cluster on GPU, time it, save the *same* events;
- 3 cluster those saved events with FastJet;
- 4 **check they agree** — else the timing is meaningless.

Both clusterers read the **same saved file**. Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.

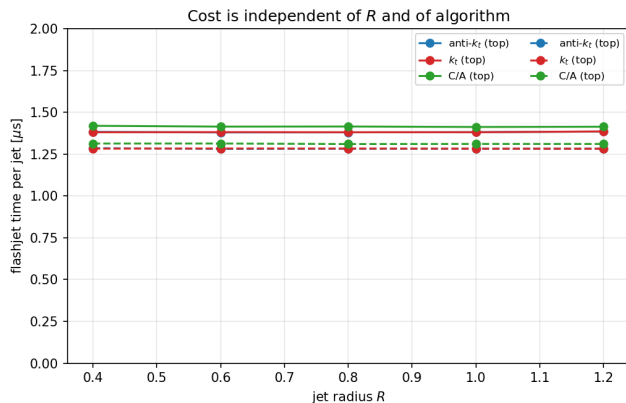
ATLAS Top Tagging Open Data — anti- k_t $R = 1.0$ 

Log scale — each gridline is $10\times$. flashjet stays nearly flat while both FastJet interfaces climb steeply with multiplicity: **0.35–2.7 μs per jet** against **17–155 μs** for vectorised FastJet, and **110–620 μs per jet** one at a time. Benchmarked on an NVIDIA V100-class GPU.

Speedup vs multiplicity — $R = 1.0$ (all points 100% n_{jets} agreement)



All three algorithms, both samples, $R = 1.0$. **39–99 $\times$** over vectorised FastJet across the grid (median **64 $\times$**), and every point has **100 %** n_{jets} agreement. Busier jets $\rightarrow$ bigger win — the regime that matters for boosted-object physics.



Across $R = 0.4 \rightarrow 1.2$: a **0.3 % spread** over a $3\times$ range in R .

R changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti- k_t , k_t and C/A agree to within $\sim 4\%$ at every multiplicity.

k_t and C/A cost the same as anti- k_t .

Downstream: the clustering tree as tagger input

Does the tree help a tagger? — setup

The question. Gouskos & Maier [PLuM, 2605.26821] report that explicit Lund-plane splittings help a transformer tagger. We test that on a **full 10-class** training.

Model. Particle Transformer [2202.03772] — 8 encoder + 2 class-attention blocks, 8 heads, embed 128. **2.143 M** params baseline, **2.193 M** with the feature block.

Data. JetClass, 10 classes, **20.05 M** test jets (~ 2.0 M/class), 128 particles/jet. All arms read **identical** shards; features computed live.

Ranger, lr 10^{-3} , batch 512, 10^6 iterations, checkpoint every 20k. Uncertainties: bootstrap of the test set, 16–84th percentile.

The feature arms

- **PLuM** — 48 k_t -ordered Lund splittings, $[\ln k_t, \ln 1/\Delta R, \ln z]$, embedded and **concatenated** as extra tokens (176 vs 128)
- **subject** — n -body subject mass, p_T fraction, n_{const}
- **CA5** — per-particle C/A branch-point features

Cost (wall clock per step)

PLuM	1.59×
subject	1.42×
CA5	1.22×

One seed per arm — see the caveat on the next slide.

Does the tree help a tagger? — result

Aggregate test accuracy (20.05 M jets)

	40k	100k	1M
baseline	83.431	84.632	86.212
PLuM	83.428	84.697	86.206
Δ	-0.003	+0.065	-0.006

AUC agrees to $\sim 10^{-4}$ at every stage.

Rejection ratio @90 % eff. (PLuM/baseline)

	40k	100k	1M
Hbb	1.009	1.037	0.983
Hcc	1.027	1.027	1.001
Hgg	1.009	1.022	1.000
H4q	1.037	1.012	1.005
Zqq	1.011	1.002	1.000
Wqq	1.036	1.001	0.999
Tbqq	1.095	1.026	0.995

What we see

An **early gain in every class** — Tbqq 1.095 ± 0.016 ($\sim 6\sigma$), Wqq 1.036 ± 0.004 ($\sim 9\sigma$) — that **decays to nothing by 1 M**. Wqq has the *fewest* splittings yet gains as much as anything: the effect is **not *b*-specific**.

At convergence **all three arms are null-or-worse**: plain training wins per GPU-hour.

What we do *not* claim

- **No mechanism.** *b*-specificity is excluded by the data; a substructure-complexity correlation was tested, not established.
- **One seed per arm.** A second PLuM seed gave a spread *larger* than the effect — the main weakness.
- We reproduce their Hbb at **100k** (1.037), **not at convergence** (0.983).

- 1 **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjects, soft drop, Lund coordinates — comes free as array reads.
- 2 **Gives the same jets as FastJet.** 150/150 grid points at 100 % n_{jets} agreement, across anti- k_t/k_t /C-A, $R = 0.4\text{--}1.2$ and 24–123 constituents per jet.
- 3 **Reproduces the experiment's own reconstructed jets** exactly, at ~ 600 clusters per event.
- 4 **39–99 $\times$ faster** than vectorised FastJet, and the advantage *grows* with jet complexity. Cost is independent of R and of algorithm.
- 5 Still a **prototype**, with meaningful headroom left.

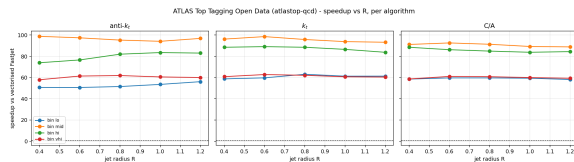
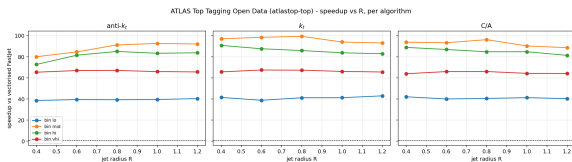
Thank you

Questions?

Benchmarks: ATLAS Open Data (CC0).

Physics closures: analytic predictions on toy showers.

Backup

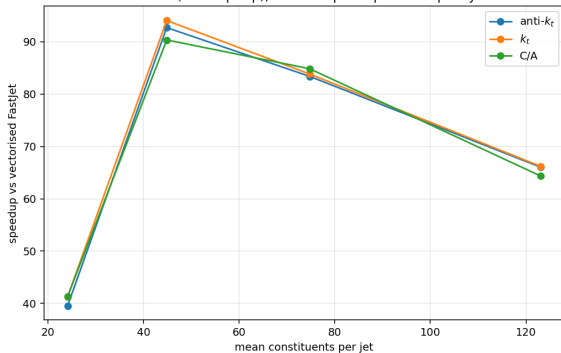


boosted top

Flat in R for all three algorithms; ordering is by multiplicity bin.

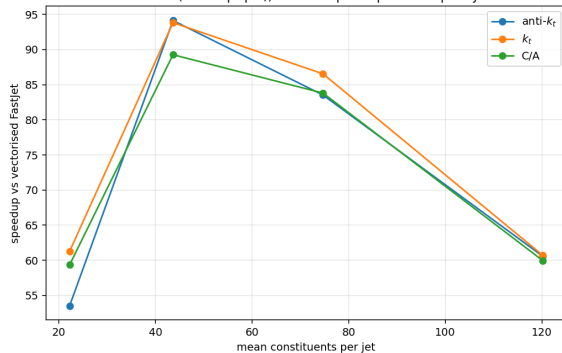
QCD (light q/g)

ATLAS (atlastop-top), R=1.0 - speedup vs multiplicity



boosted top

ATLAS (atlastop-qcd), R=1.0 - speedup vs multiplicity



QCD (light q/g)

Backup — the numbers

sample	alg	$\langle n \rangle$	flashjet [$\mu\text{s}/\text{jet}$]	FastJet vec. [$\mu\text{s}/\text{jet}$]	speedup
top	anti- k_t	24.1	0.427	16.5	39×
top	anti- k_t	44.8	0.464	43.0	93×
top	C/A	74.7	1.000	84.8	85×
top	C/A	123.1	2.565	165.1	64×
qcd	anti- k_t	22.2	0.346	17.5	51×
qcd	anti- k_t	43.6	0.463	45.7	99×
qcd	anti- k_t	74.7	1.215	89.8	74×
qcd	anti- k_t	120.2	2.673	154.9	58×

39–99× over vectorised FastJet (median 64×), with 100 % jet-count agreement everywhere. Benchmarked on an **NVIDIA V100-class** GPU.

$R = 1.0$ rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

Backup — where the time goes

Nsight Systems — share of GPU time

kernel	% GPU	avg/call
_cluster_large_kernel	97.6	5.47 ms
_decode_kernel	1.5	82 μ s
torch glue	< 1	—

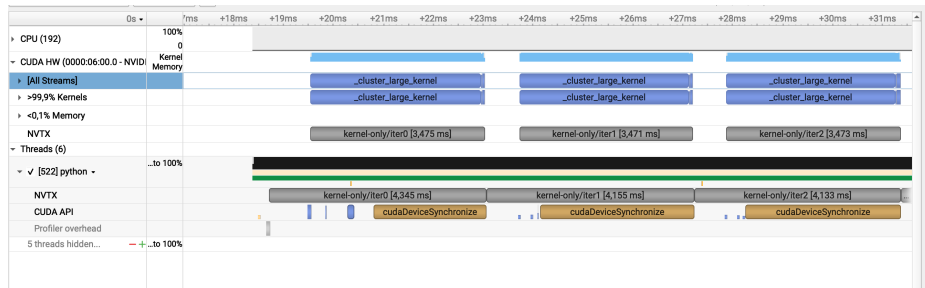
No hidden overhead — essentially all the time is real clustering work, and the memory row of the timeline is empty.

Nsight Compute — hardware counters

metric	value
DRAM throughput	0.02 %
Compute (SM) throughput	15.4 %
Registers / thread	168
Theoretical occupancy	18.75 %
Achieved occupancy	6.25 %
Waves per SM	0.32

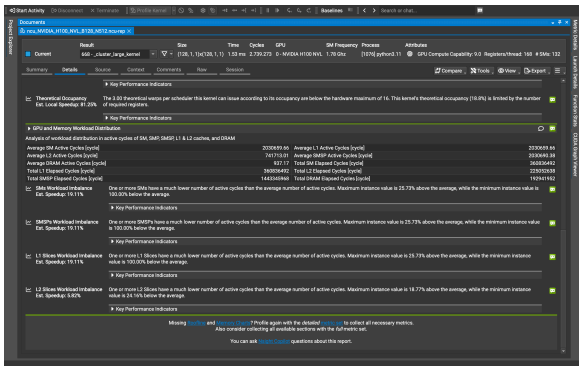
Not memory- or compute-bound — the GPU is simply **not filled**.

Backup — the GPU timeline



Nsight Systems, three consecutive iterations. Each block is one clustering call; the tool labels the rows “>99.9 % Kernels” and “<0.1 % Memory”. The kernels run back to back with nothing in between and the memory row is empty — no host/device shuffling while clustering runs. That is what makes the speedup real rather than an artefact of where the timer was started.

Backup — the profiler names the bottleneck



The tool's own speedup estimates:

finding est. speedup

Theoretical occupancy (registers) 81 %

Achieved occupancy 67 %

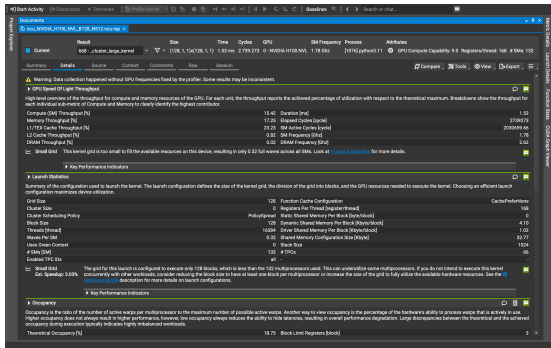
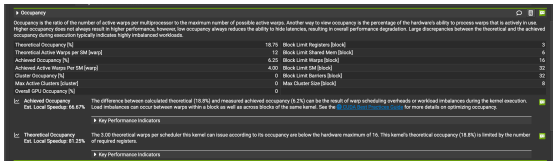
SM workload imbalance 19 %

"This kernel grid is too small to fill the available resources on this device, resulting in only 0.32 full waves across all SMs."

Grid size **128 blocks** on a **132-SM** device — fewer blocks than the GPU has processors.

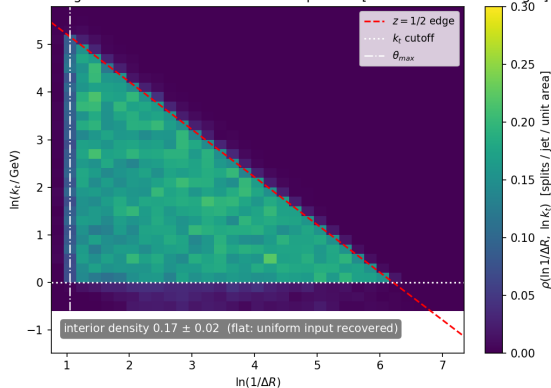
~**1.7–1.8×** more available from launch configuration and register budget — **not** from the algorithm.

Backup — occupancy detail



Register pressure (168/thread) caps theoretical occupancy at 18.75 %; achieved is 6.25 %. With 0.32 waves/SM the kernel does not fill the GPU even once, while DRAM throughput is 0.02 % — this is parallelism starvation, not a bandwidth problem. Both are configuration inherited from smaller GPUs, not properties of the algorithm.

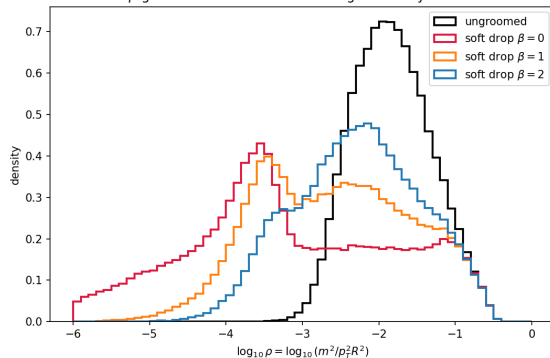
primary Lund plane of the toy shower (C/A, R=0.4):
m-in-triangle emissions come back as a flat plateau [cf. arXiv:1807.04758 Fig 2]



The Lund plane is populated **uniformly** inside the kinematic **triangle** — the leading-log expectation.

Independent NumPy tree-walks pin every decoder, and the d -channel of the Lund output ties *exactly* to the splitting scales computed separately.

Soft-drop β -ordering on REAL QCD jets (164292 jets, $z_{cut} = 0.1$)
smaller β grooms harder — same ordering as the toy-shower closure



The soft-drop β -family orders as predicted: larger β grooms less.

Backup — Input A: the toy fat jets

Purpose. QCD vs W is a *known* truth, so any observable that fails to separate them — or lands off the analytic curve — is a real bug.

Two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:

- **QCD-like:** one hard collinear core (92 % of p_T , $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8 %, $\sigma = 0.30$)
- **W-like:** two hard prongs with pair mass $m = \sqrt{z(1-z)} p_T$, $\Delta R = 80.4$ GeV, $z \in [0.30, 0.45]$, random orientation, + 6 % soft contamination

Each “spray” is n massive pions with exponential p_T fractions:

```
frac = rng.exponential(1., n); frac /= frac.sum()
pt    = pt_total * frac
y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
mt     = np.sqrt(M_PI**2 + pt**2)
```

1500 events per sample. `flashjet` has **no event generator** — it clusters four-vectors; these toys live in the plot scripts, outside the library.

Backup — Input B: the toy parton shower

A **primary, fixed-coupling, leading-log** shower: emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta, \ln k_t)$ area, off a hard spine.

$$\theta = e^{-u}, \quad k_t = e^v, \quad z = \frac{k_t}{p_{T0} \theta}$$

keep $v \leq \ln(p_{T0}/2) - u$ (i.e. $z \leq \frac{1}{2}$), $k_t > k_t^{\min}$

Why this is the right test

The generator's density is *uniform in the triangle by construction*, so the leading-log predictions — the $1/z$ form of z_g , the β -ordering of the groomed mass — are unambiguous. Landing on them tests the **decoders**, not the shower.

Jet areas use the anti- k_t -paper construction: one hard event plus a dense grid of **infinitely soft ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area.

$\bar{\alpha} = 0.25$, $p_{T0} = 1$ TeV, $R = 0.4$, 20 000 showers; fixed seed, fully re-runnable.

Backup — the validation ladder

rung	input	reference	result
unit tests	—	independent NumPy tree-walks	129 pass, incl. GPU paths
anti- k_t jet areas	toy B	the anti- k_t paper's own construction	rigid cones, $\rightarrow \pi R^2$
k_t substructure	toy A	known 2-prong vs QCD truth	clean separation
soft-drop z_g	toy B	leading-log $1/z$ prediction	on the curve
groomed mass vs β	toy B	Soft Drop Figs. 3–4	ordering reproduced
Lund plane	toy A	primary-Lund construction	hard split where predicted
jet-by-jet kinematics	real C	FastJet, same events	150/150 points, 100 % n_{jets}
full-event clustering	real C	the experiment's own jets	100 % match, ~ 600 clusters/ev

Each rung is closed against something independent — an analytic prediction, a separate NumPy implementation, or FastJet itself. No rung is validated against another part of flashjet.

- H. Qu, C. Li, S. Qian, *Particle Transformer for Jet Tagging*, arXiv:**2202.03772**. (also the JetClass dataset; Zenodo 10.5281/zenodo.6619768)
- L. Gouskos, B. Maier, *Particle-Lund Multimodality in Jet Taggers*, arXiv:**2605.26821**.
- H. Qu, L. Gouskos, *Jet tagging via particle clouds* (ParticleNet), Phys. Rev. D **101** (2020) 056019, arXiv:1902.08570.
- F. A. Dreyer, G. P. Salam, G. Soyez, *The Lund Jet Plane*, JHEP **12** (2018) 064, arXiv:1807.04758.
- F. A. Dreyer, H. Qu, *Jet tagging in the Lund plane with graph networks*, JHEP **03** (2021) 052, arXiv:2012.08526.
- A. J. Larkoski, S. Marzani, G. Soyez, J. Thaler, *Soft Drop*, JHEP **05** (2014) 146, arXiv:1402.2657.
- M. Cacciari, G. P. Salam, G. Soyez, *The anti- k_t jet clustering algorithm*, JHEP **04** (2008) 063, arXiv:0802.1189; *FastJet user manual*, Eur. Phys. J. C **72** (2012) 1896, arXiv:1111.6097.
- S. Catani, Yu. L. Dokshitzer, M. H. Seymour, B. R. Webber, Nucl. Phys. B **406** (1993) 187 (k_t); Yu. L. Dokshitzer *et al.*, hep-ph/9707323 and M. Wobisch, T. Wengler, hep-ph/9907280 (Cambridge/Aachen).
- J. de Favereau *et al.*, *DELPHES 3*, JHEP **02** (2014) 057, arXiv:1307.6346.

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

dataset	used for
ATLAS Top Tagging Open Data	timing + FastJet agreement
2020 ATLAS Jet Reconstruction	closure vs the experiment's own jets

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.